

13.0 Milestone Assessment: Network Addressing Schemes

Overview

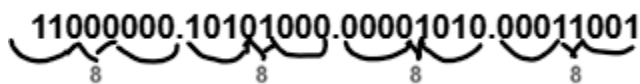
Network addressing provides the structure that allows devices to identify one another and communicate across local and wide area networks. A properly designed addressing scheme must account for the number of devices on the network, network segmentation, routing, security, scalability, and the services used to automatically or manually configure network hosts.

Two primary Internet Protocol addressing systems are used today: Internet Protocol version 4 (IPv4) and Internet Protocol version 6 (IPv6). IPv4 remains widely deployed, while IPv6 provides a much larger address space designed to support the continued growth of interconnected devices, including the Internet of Things (IoT). Subnetting and Classless Inter-Domain Routing (CIDR) provide additional methods for efficiently organizing address space and controlling how networks are routed.

IPv4 Addressing

IPv4 uses a 32-bit address divided into four, 8-bit octets. Each octet is represented as a decimal number from 0 through 255. For example: 192.168.10.25. In binary, an IPv4 address contains 32 individual bits, or 4 sets of 8 bits (four, 8-bit octets):

11000000.10101000.00001010.00011001



The subnet mask or CIDR prefix identifies which bits represent the network and which are available for hosts. For example, 192.168.10.25/24 uses the first 24 bits for the network.

IPv4 provides approximately 4.3 billion addresses, which became insufficient as the number of Internet-connected devices increased. Private IPv4 ranges and Network Address Translation (NAT) have helped conserve public addresses, but they do not increase the actual IPv4 address space.

IPv6 addresses this limitation by using 128-bit hexadecimal addresses, such as 2001:db8:10:1::25/64. IPv6 provides approximately 3.4×10^{38} addresses, allowing an extremely large number of devices to be uniquely addressed. IPv6 also eliminates broadcast

communication and supports address types such as global unicast, link-local, and multicast. IPv6 devices may be configured statically or through SLAAC and DHCPv6.

Understanding both protocols is important because modern networks may operate IPv4 and IPv6 together using a dual-stack configuration.

IPv4 and IPv6 Comparison

Characteristic	IPv4	IPv6
Address length	32 bits	128 bits
Representation	Decimal	Hexadecimal
Example	192.168.10.25	2001:db8::25
Typical LAN prefix	/24	/64
Address capacity	~4.3 billion	$\sim 3.4 \times 10^{38}$
Broadcast	Supported	Not used
Multicast	Supported	Supported
NAT	Common	Generally not required for address conservation
Address configuration	Static or DHCP	Static, SLAAC, and/or DHCPv6

IPv4 and IPv6 can coexist through technologies such as dual-stack networking while organizations transition between the protocols.

Subnetting

Subnetting divides a larger network into smaller, manageable networks. Instead of placing every device into the same subnet, the organization could divide the address space into smaller networks for different departments or functions. For example, a **192.168.10.0/24** network could be divided into four **/26** subnets:

Example:

Network	Subnet
Administration	192.168.10.0/26
Sales	192.168.10.64/26
Operations	192.168.10.128/26
Guest Network	192.168.10.192/26

Each /26 contains 64 total addresses and 62 traditionally usable IPv4 host addresses. This provides a more structured network than placing every device into one large broadcast domain.

Subnetting improves organization, performance, security, and troubleshooting. Smaller subnets reduce broadcast domains, while separating departments, servers, management systems, and guest devices allows routers, firewalls, VLANs, and ACLs to control communication between network segments.

For example, guest wireless devices can be placed on a separate subnet and prevented from accessing internal servers while still being allowed Internet access.

Classless Inter-Domain Routing and Address Conservation (CIDR)

CIDR represents the network prefix by identifying how many bits belong to the network portion of an IP address. For example: 192.168.1.0/24. The /24 indicates that 24 bits identify the network. CIDR replaced the limitations of traditional classful IPv4 addressing by allowing networks to be allocated according to actual size requirements instead of being restricted to fixed Class A, Class B, or Class C boundaries.

Common IPv4 prefixes include:

CIDR	Subnet Mask	Total Addresses	Traditional Usable Hosts	Subnets
/24	255.255.255.0	256	254	1
/25	255.255.255.128	128	126	2
/26	255.255.255.192	64	62	4

CIDR	Subnet Mask	Total Addresses	Traditional Usable Hosts	Subnets
/27	255.255.255.224	32	30	8
/28	255.255.255.240	16	14	16
/29	255.255.255.248	8	6	32
/30	255.255.255.252	4	2	64

CIDR replaced the limitations of traditional classful addressing by allowing networks to be sized according to actual requirements. A network requiring approximately 50 host addresses, for example, can use a /26 instead of allocating an entire /24.

CIDR also supports route aggregation, or route summarization. Multiple contiguous networks can sometimes be represented by a single summarized route, reducing the number of routing-table entries routers must maintain. CIDR therefore helps conserve IPv4 address space while also improving routing efficiency. This reduces the number of individual entries routers must maintain and can simplify routing tables.

CIDR serves two important purposes:

1. More efficient allocation of IPv4 address space.
2. More efficient routing through route aggregation.

Supporting Network Services

IP addressing works with several additional network services.

1. MAC addresses operate at Layer 2 and identify network interfaces for local Ethernet communication. IP addresses provide logical Layer 3 addressing for communication between networks, while MAC addresses are used to deliver frames across the local network.
2. DHCP automatically provides clients with configuration information such as an IP address, subnet mask, default gateway, and DNS server. In IPv4, the DHCP process can be remembered as DORA: Discover, Offer, Request, Acknowledge.
3. DNS translates host and domain names into IP addresses. Common records include **A** for IPv4, **AAAA** for IPv6, **CNAME** for aliases, **MX** for mail servers, and **PTR** for reverse lookups. A device can have valid IP connectivity but still appear unable to reach a resource if DNS resolution fails.
4. Static addressing is useful for infrastructure that must remain consistently reachable, including routers, servers, firewalls, printers, switch management interfaces, and other

network services. Client devices are more commonly configured dynamically through DHCP.

Practical Network Addressing Example

Consider a small organization with separate employee, server, and guest networks. The organization could use:

Purpose	Network	Addressing
Employees	192.168.10.0/24	DHCP
Servers	192.168.20.0/24	Static/reserved
Guest Wi-Fi	192.168.30.0/24	DHCP
Network Management	192.168.40.0/24	Static/reserved
IPv6 Employees	2001:db8:10:10::/64	SLAAC/DHCPv6
IPv6 Servers	2001:db8:10:20::/64	Static/reserved as appropriate

This design separates systems according to purpose while providing predictable addressing for infrastructure and automatic configuration for client devices. Firewall rules or Access Control Lists (ACLs) could prevent the guest subnet from reaching internal server and management networks while still allowing Internet access.

This demonstrates how addressing, subnetting, DHCP, DNS, routing, and security work together rather than functioning as isolated technologies.

Network+ Application

These concepts work together when configuring and troubleshooting a network. A technician may need to determine whether two devices are on the same subnet, calculate the number of available hosts, identify an incorrect subnet mask or default gateway, recognize IPv4 and IPv6 address types, verify DHCP configuration, troubleshoot DNS resolution, or determine whether a device should use static or dynamic addressing.

For example, a network could use separate subnets for employees, servers, guests, and network management. Employee and guest devices could receive addresses through DHCP, while servers and infrastructure use static addresses or reservations. IPv6 /64 networks could

be assigned alongside IPv4 networks, and security policies could control communication between the different segments.

Commands such as `ipconfig` or `ip`, `ping`, `tracert/traceroute`, `nslookup`, and `arp` can then help determine whether a communication problem involves IP addressing, routing, DNS, or Layer 2 connectivity.

Conclusion

Network addressing schemes form the foundation of TCP/IP communication. IPv4 provides the traditional 32-bit addressing system still widely used today, while IPv6 expands the address space to 128 bits to support continued network growth. Subnetting divides networks into smaller logical segments, improving organization and supporting security, performance, and troubleshooting. CIDR provides flexible prefix lengths that improve IPv4 address utilization and allow route aggregation to make routing more efficient.

MAC addressing, DHCP, DNS, and static configuration complement IP addressing at different layers and stages of network communication. Together, these technologies allow administrators to build networks in which devices can be uniquely identified, automatically or manually configured, efficiently routed, logically segmented, and reliably located.

Understanding how these technologies interact is essential not only for the Network+ exam but also for designing, administering, securing, and troubleshooting real-world networks.

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